RPMGLOBAL

AMT - Strategy Optimisation

Training Manual

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RPM SOFTWARE – ASSET MANAGEMENT

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Preface

The training workshops are developed to supplement instructor-led training. This guide is constructed in a way that teaches students the key functionality of the application through practical hands-on workshops. Each workshop details tasks that the student is required to complete. The workshops build on each other to help ensure a comprehensive understanding of the application is achieved at the completion of the course. Space is provided at the end of each workshop for learning checks and student notes.

Symbol Library



Note

A note for participants about the task.



Important Reminder

An important reminder for the participant about the task.



Learning Check

Check your understanding on the workshop and ask any related questions.

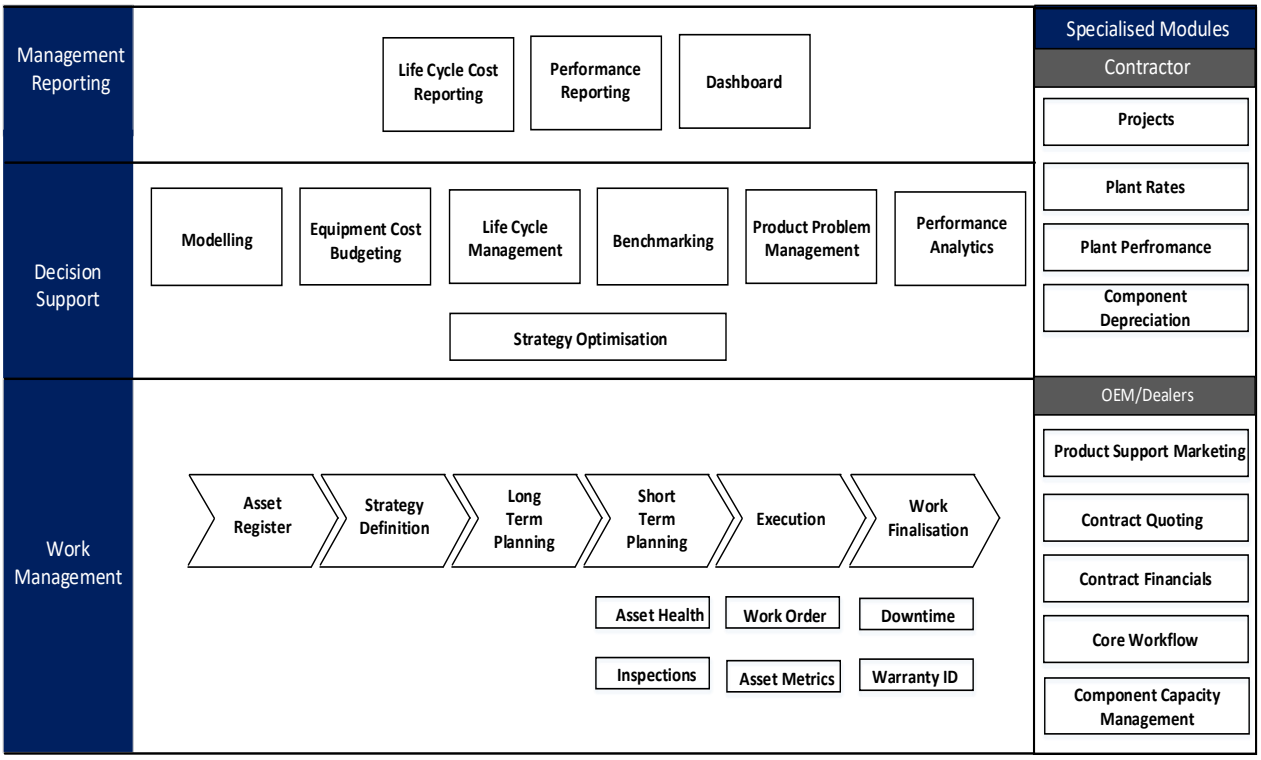
Course Outline

Prerequisites	Getting Around AMT, AMT Key Concepts
Target Audience	Maintenance Manager, Strategist, Reliability Engineers, Contract Manager (OEM), Sales Rep (OEM)
Nominal Duration	
Course Objectives	The objectives of this course are: • To optimise equipment performance and minimise operating costs. • To drive continuous improvement. • To ensure AMT projections are accurate.
Learning Outcomes	At the end of the course, participants will have learned to: • Identify Opportunities and Risks • Identify components that are not performing to plan o Determine changes required to the strategy to mitigate risks o Understand PCV o Understand RVIC o Understand ACI and how to apply it
	• Test solutions using alternate projections (What if) o Create alternate projections o Test impact of making changes o Update the current projection with changes • Fine tune long term planning o Identify risk and opportunity components o Modify long term plan to maximise returns and minimise risk
Assessment Strategy	The completion of workshop tasks.
AMT version	8.26

Introduction to AMT

AMT is a comprehensive asset management system consisting of integrated modules as depicted below.

The AMT Strategy Optimisation functionality provides tools and techniques to assist in the implementation of a proactive, risk-based maintenance approach. AMT strategy promotes a holistic integrated approach to maintenance for better productivity, effectiveness and profitability.



FUNCTIONALITY OVERVIEW

This training manual describes the process of optimising the maintenance strategy for an asset.

The maintenance strategy drives life cycle costs, component schedules, parts forecasts, labour forecasts, maintenance plans and component risks and opportunities. In AMT, maintenance strategies are stored in **projections**.

The strategy process consists of a number of related and parallel activities that support the maintenance effort. The core strategy processes include:

- Create the initial maintenance strategy.
- Validate data.
- Identify opportunities.
- Investigate.
- Test solutions.
- Action.
- Reporting.

LOGIN AND NAVIGATION

For information on AMT login and navigation, refer to the **AMT Getting Around** training manual.

TERMINOLOGY

For information on AMT terminology, refer to the **AMT Glossary** training manual and in the online help.

SYSTEM CONFIGURATION & SETTINGS (LINK TO EXTERNAL STRATEGY TASKS – SAP)

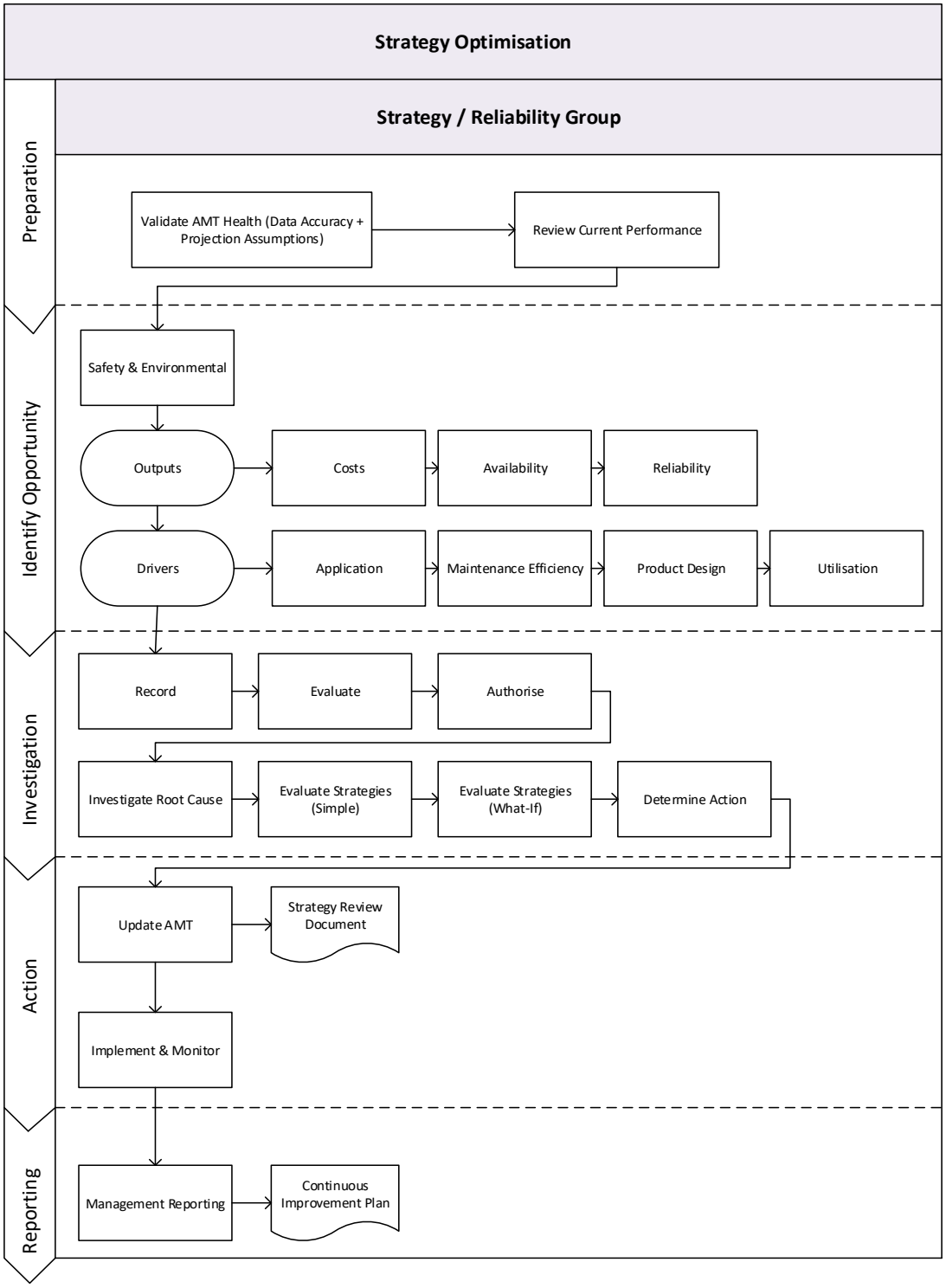
For some installations of AMT, the current maintenance strategy (represented in AMT as strategy tasks linked to an equipment current projection), is stored in an external system. An automated interface is used to send maintenance plans and job costs from the external system to AMT, typically daily.

In this scenario, the maintenance strategy is maintained in the external system and AMT is used to forecast and analyse the current maintenance strategy. Changes to the maintenance strategy are made in the external system, which interfaces the updates back to AMT.

The link between the strategy task in AMT and the equivalent strategy task in the external system is defined in the **External Identifier** field in each AMT strategy task. The value in the field is typically formatted to be a concatenation of [EquipmentID + ComponentCode + ModifierCode + ExternalSystemStrategyTaskID], which guarantees uniqueness.

Processes

PROCESS DIAGRAM



PROCESS INPUTS

The following inputs are required for this process:

Input	Description
Equipment current and future usages	Enter current equipment usages into AMT to ensure the next due component hours are correct and to update usage profiles if necessary.
Understanding of application	To ensure the strategy is relevant, you must know about the future application of the machine. If the loader has moved from overburden to ore, how does this affect the component strategy? If a pump now no longer has a 100% standby, how does this affect the maintenance strategy?
Knowledge of component improvements or engineering changes	In order to accurately forecast future occurrences, you must understand changes to the design of the components that affect its ultimate life (such as product improvement bulletins and local modifications).
Equipment disposal dates	The strategy needs to reflect the disposal date of the equipment. If an asset is to be retired in 6 months, a different approach to strategy may be required. Conversely, the strategy may alter the equipment disposal decision.
Component health data	External conditioning monitoring inputs that help assess the expected performance of the component such as oil sample results, Caterpillar VIMS data, vibration measurements, visual inspections and so on.

PROCESS OUTPUTS

The following outputs can be expected from this process:

1. Documented continuous improvement initiatives.
2. Documented maintenance strategy.
3. Accurate productivity, cost and resource projections.

Training Workshops

1. Strategy Definition

The strategy for an asset is defined in the current projection. The projection contains:

- High cost and risk items that need to be forecast and tracked (defined in AMT as strategy tasks).
- Life expectancy of these strategy tasks.
- Cost to replace or repair these strategy tasks.
- Expected utilisation of the asset.

The output of the process is a full life cycle cost projection for the equipment (see the example below):

Projection Summary

Hierarchy : [Equipment Plan *: TR01] [Projection Type *: Current]
Additional Filters : [Escalation:Normal] [Currency *: USD US Dollars]
Date Range : [Entire Plan Term]
Analyse By : [Component Code]

Component Description	UOM	Cost To Date	First	Freq	Future Unit Cost				Future Cost		Total Cost	Cost Rate
					Parts	Labour	Misc	Total	Occs	Total		
1000.00.CC.0 ENGINE	Eng - Hrs	1,110,946	18,000	22,000	166,789	35,300	19,420	221,509	1	221,509	1,332,456	
1000.00.RR.0 ENGINE	Eng - Hrs	43,043	2,200	2,200	4,660	1,029	555	6,243	6	37,459	80,501	
1052.LF.CC.0 TURBOCHARGER	Eng - Hrs	12,273	9,000	9,000	6,237	2,585	0	8,822	1	8,822	21,094	
1052.LR.CC.0 TURBOCHARGER	Eng - Hrs	12,247	9,000	9,000	6,237	2,585	0	8,822	1	8,822	21,068	
1052.RF.CC.0 TURBOCHARGER	Eng - Hrs	11,429	9,000	9,000	6,237	2,585	0	8,822	1	8,822	20,250	
1052.RR.CC.0 TURBOCHARGER	Eng - Hrs	19,262	9,000	9,000	6,237	2,585	0	8,822	1	8,822	28,083	
1054.IN.PM.0 AIR FILTER ELEMENT	Eng - Hrs	1,357,401	1,000	1,000	587	38	0	624	13	8,116	1,365,517	
1054.OU.PM.0 AIR FILTER ELEMENT	Eng - Hrs	4,497,016	250	250	479	38	0	517	51	26,353	4,523,369	
1062.00.GR.0 MUFFLER	Eng - Hrs	1,823,911	18,000	18,000	16,177	1,551	0	17,727	0	0	1,823,911	
1105.00.PM.0 Perform Valve & Injector Adjustment	Eng - Hrs	758,385	0	4,000	0	2,000	0	2,000	3	6,000	764,385	
1250.00.CC.0 FUEL SYSTEM	Eng - Hrs	354,385	9,000	9,000	22,642	2,068	0	24,709	1	24,709	379,095	
1353.00.CC.0 RADIATOR	Eng - Hrs	22,831	18,000	18,000	15,300	4,135	0	19,435	1	19,435	42,266	
1355.00.CC.0 Thermostats	Eng - Hrs	461	6,000	6,000	274	500	0	774	3	2,321	2,782	

2. Life Cycle Analysis

A number of reports in the Strategy module analyse historical and projected costs based on a single projection. Other reports compare the difference between two projections. A key concept in a number of these reports is projected cost variance (PCV).

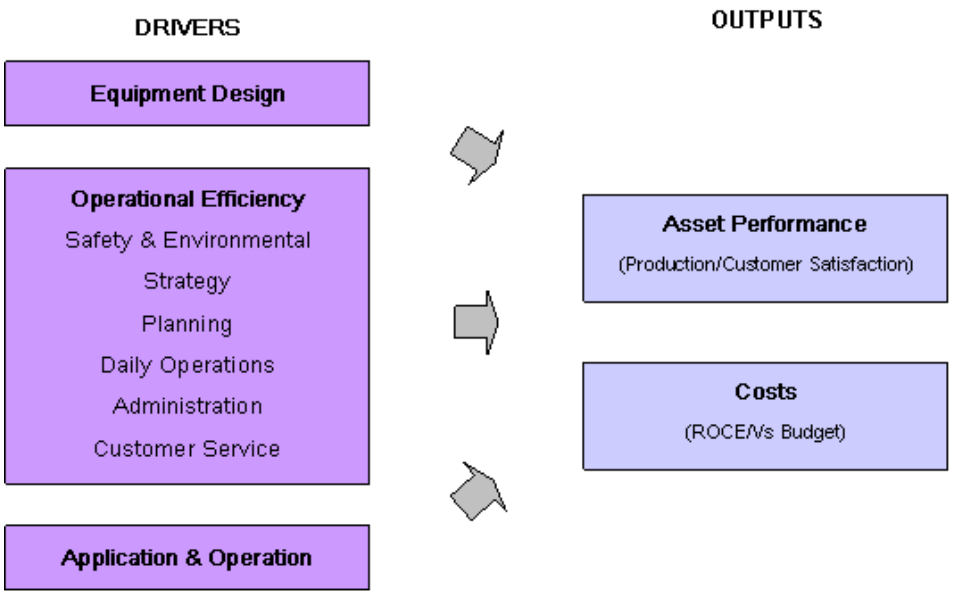
To have confidence in both types of analysis, it is critical that the health of AMT's data is at an acceptable level. The primary health driver in this context is the quality and accuracy of transactional data (that is, work order related data) and projection assumptions (estimate cost and estimate life). Therefore, to understand the current position (as reported by AMT) it is necessary to validate data in AMT, using the AMT Health Assessment process.

For information on performing health assessments, refer to the **AMT Health Assessment** training manual.

3. High Level Performance Review

The objective of this step is to get a high-level understanding of the asset performance, so you know where to focus your efforts in identifying opportunities. For example, if the costs are on budget but there is low availability, then you focus on the downtime analysis.

In the illustration below, it is important to distinguish between outputs and drivers. Continuous improvement initiatives can only focus on a driver that results in increased availability, lower costs, and so on.



4. Identify Opportunities

Outlined below is a suggested approach to identify opportunities. You will need to tailor this approach depending on the type of asset, the criticality of the asset and the performance level of the asset. For example, more time would be spent investigating poor performance on a critical loader than a support grader.

The objective of AMT is to highlight potential issues so that you can prioritise maintenance. AMT does not generally highlight the root cause of an issue. For example, AMT highlights there is an overheating problem on motor, but you need to determine what the cause of the overheating is.

The suggested approach is to evaluate both the outputs and the drivers.

OUTPUTS

Area	Report	Notes	Drill down tips
Costs			
	High Cost Areas	Objective is to identify Identify high cost items. Even if performance is in line with expectations one needs to ask the question "Can we improve the strategy?"	
	Cost Analysis - Life-Cycle	Run the report for the "life-cycle" or a shorter period (ie a year). Pie chart, Analyse by Component will show top 20 component cost areas	Strategy Task Analyzer - will show the average and projected cost and frequency and the PCV.
	Cost v Accounting Budget	Applicable for users of AMT Budgeting Module. Allows users to compare actual costs against accounting budgets and analyse variances (See Budgeting Procedure)	
	Component Performance	Specific analysis of the performance of major components. Focus on identifying the different types of failure that are occuring and looking at ways to prevent them. For those using the CBM Module there is more detailed reliability anayses - see CBM Procedure	
	Technical Performance Analysis	Run for each major component in a Fleet, analyse by Occurrence type (Failure Mode). Shows the average life/cost for each Failure Mode	Run the same report again but add additional "Analyse by" criteria such as Cause Technical Performance Details report shows the technical details (Cause, Symptom, Part # etc) for each changeout Component History report shows the specific workorder history and shows "Partial Life" and "Swing" components Component Life Distribution (analyse by Occurrence Type) or Cause
	CBM - Reliability Analyses	Applicable for users of AMT CBM Module. Allows users to trend condition monitoring data leading to component failures to identify failure indicators and refine condition monitoring strategies. (See Budgeting Procedure)	
	Unassigned Costs	Evaluate the level of unassigned costs and drill down to the cause	
	Cost Analysis - To date	Run the Pie Chart and analyse by Task Type. This shows the unassigned costs in relation to all costs to enable users to evaluate if the level is reasonable.	
	Cost per UOM - By Date	Filter for Unassigned Task Type and analyse by Model/System/Sub-System for a date range. Allows users to see the trending of unassigned costs	
	Unassigned Cost Analysis	Analyse by Component Code. Will identify the components that have the highest unassigned costs	
	Component Life Risk	Shows the opportunity/risk associated with components life at the end of the equipment's life (see notes on RVIC in Key concepts document).	
	High Opportunity	Components where costs can be saved if the life is extended	
	High Risk	Components that represent a high risk of incurring additional costs if there is slight reduction in component life	

Area	Report	Notes	Drill down tips
Cost Over-runs		Identify actual and potential cost over-runs against the maintenance strategy assumptions	
	Life-Cycle Cost Summary	Applicable for those equipment with a Budget Projection. The PCV highlights projected cost over-runs from original/baseline assumptions	Strategy Task Analyzer - then click Va icon to go to the Variance Analysis report. This shows whether it is the life or cost that is driving the
	Strategy Task Analyzer	The ACI highlights actual cost history that varies from your projection assumptions. Users can sort Strategy Tasks from highest to lowest so they can quickly identify the items with the biggest impact.	
Availability			
Availability		Objective is to highlight equipment with performance issues. Depending on the site either Mechanical or Physical Availability is used.	
	Availability - Mechanical	To Trend, analyse by month. To compare equipment/fleets then analyse by equipment/fleet.	Usage analysis chart - With this calculation method, low utilisation can cause low availability High Downtime - see below
	Availability - Physical	To Trend, analyse by month. To compare equipment/fleets then analyse by equipment/fleet.	
High Downtime		Identify biggest contributors to Downtime	
	Downtime Analysis	Pie chart, analyse by component code to identify high downtime areas	Same report but drill down on high downtime areas
Downtime Var: Planned v Actual		Identify tasks that exceed the planned downtime	
	Downtime Variance Analysis	Being developed (not in AMT 8.2)	
Non-Planned		Looking to identify tasks that could have been planned (ie included in PM services)	
	Downtime Analysis	Pie chart, filter for "non-planned" and "non-Breakdowns" and analyse by component code to identify high downtime areas.	Same report but drill down on high downtime areas
Non-Maintenance Activities		Looking to identify planning or process inefficiencies that are contributing to downtime	
	Downtime Analysis	Pie chart, filter for all "non-maintenance" Activities and analyse by Activity and Variance cause.	
Reliability			
Reliability		Objective is to highlight equipment with reliability issues.	
	MTBF	To Trend, analyse by month. To compare equipment/fleets then analyse by equipment/fleet.	
Unplanned Stoppages		Identify the cause of breakdown stoppages	
	Event Analysis	Filer for Breakdowns and analyse by Component	Drill down by Cause

DRIVERS

Report	Description	Impact		
		Cost	Availability	Reliability
Backlog Ageing Backlog Work Analysis	High backlogs are one of the key maintenance "lead indicators" and can indicate a number of potential issues: 1) Increase in maintenance workload from: - a change in the application/operating practices - product problems - quality of maintenance work 2) Shortage of maintenance labour 3) Shartage of maintenance parts 4) Planning inefficiencies As backlogs increase so the risk of failures increase. It also highlights a future drop in availability	√	√	√
Planned Downtime	For both the events and Hours this indicates the level of "control" with the maintenance organisation. If the trend is down then it indicates either: - reliability problems (see Breakdown analysis below) - planning inefficiencies An analysis of "non-planned" and "non-breakdowns" is a good way to identify planning inefficiencies. Planning inefficiencies lead to increased stoppages and downtime (ie work that could have been done in a PM get done as a separate stoppage		√	√
Service Accuracy	Regular, timely and quality servicing forms the backbone of good maintenance practices. In addition, it is often a "warranty" requirement. The service accuracy report measures the efficiency of the planning process to plan and schedule services in line with the maintenance strategy. Not performing regular services can lead to breakdons and unplanned stoppages and also have a long term impact on component lives. The Service Accuracy Variance report shows the specific tasks that Note, incorrect or infrequent capture of utlisation (Usages) can also lead to poor service accuracy	√	√	√
MTFS after PM	This goes hand in hand with the Service Accuracy report and measures the quality of PM services. Poor quality PM services directly impact costs and productivity	√	√	√
Maintenance Ratio	Measures the man hours to operating hours. If it is high then it indicates either reliability issues or low skill levels. A usefull analysis is to trend this report and exclude component changeouts (remove the big spikes). If the trend is up then this indicates issues.	√		
MTTR	By itself, the MTTR (Mean Time To Repair) report does not mean a lot: - an increasing MTTR could indicate many things - better planning (fewer and longer stoppages) or poor skill levels (tasks take longer) Consequently it is important to view this report in a context: - With MTBS - if MTBS is increasing then you would expect MTTR to increase. If however MTBS is constant and MTTR increases then this indicates issues (poor skill levels) - Filtered only for Breakdowns and non-changeouts (remove the spikes). This indicates the average time to repair the smaller general breakdowns and is a good indicator of response times (availability of people) and skill levels	√	√	√
Redo Analysis	This is not a KPI report but a good report to view if any of the other reports are indicating potential work quality issues	√	√	√

Report	Description	Impact		
		Cost	Availability	Reliability
Breakdown Analyses	High levels of breakdowns can indicate a number of issues. Use the different filters and analyse by options to trend and then drill down to identify the cause	√	√	√
Labour Hours Analysis	Analysis where time is being spent (particularly of breakdown labour) to identify product problems	√	√	√
Labour Variance Analysis	Particularly for Strategy tasks, shows actual average labour hours against the target. Identifies: 1) Incorrect targets (ie in Standard Jobs) 2) Inefficiencies due to low skill levels or poor task instructions	√	√	√

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

5. Investigation

The investigation process varies depending on the issues identified. If the issue is specific to an equipment model (that is, a product problem), use **EM Issues** to document the resolution process.

If the issue is process related and not specific to an equipment model (such as a shortage of skilled mechanics), it should be registered in some other continuous improvement log.

The basic investigation approach is:

1. **Record** - Document the issue, describing the symptoms and impact.
2. **Evaluate** - To prioritise the issue, you need to evaluate the impact. In particular, the estimated impact on production and costs.
3. **Authorise** - Based on the above analysis and depending on the internal processes, authorisation may be required before resources are assigned to investigate the root cause.
4. **Investigate Root Cause** - The steps taken will depend on the issue identified.
5. **Evaluate Strategies (simple)** - After the root cause is identified, a number of corrective options may need to be evaluated. Evaluate these options in terms of impact on productivity, cost and resource requirements.
6. **Evaluate Strategies ("what-if")** - If a more detailed analysis of different strategies is required, use the what-if function in AMT.
7. **Determine Action** - Determine the optimal corrective action to take.

6. Action

As with the investigation, the action steps vary depending on the issue and actions required. If the action is related to an asset, the first step is to update AMT with the new strategy. This involves:

- Updating the strategy tasks (frequencies, costs, standard jobs, workscopes).
- Documenting the reasons for the change in strategy (see the **Strategy** tab in the **Strategy Task**).
- If applicable, updating the component structure (failure modes, probabilities, and so on).

The **Maintenance Strategy Review** report in AMT provides a detailed summary of the maintenance strategy for an asset.

If one-off corrective tasks are required, create them directly in **Daily Planning**. Alternatively, they can be created from **EM Issues**.

For more information on EM Issues, refer to the **AMT EM Issues** training manual.

The corrective action needs to be monitored to ensure it is carried out and achieves the desired result.

7. Analysis of Projection Assumptions

Projection assumptions are the parameters that determine at what points in time a strategy task is forecast, and how much it is estimated to cost per occurrence. These parameters are stored in the **Scheduling** and **Job Details** tabs within strategy tasks.

The Strategy module contains functionality to help validate the accuracy of these parameters (that is, to check whether the assumptions and estimations are reasonable). **Strategy Task Analyser** is the key function in this area.

Use the Strategy Task Analyser to evaluate the strategy task assumptions against actual history. On the **Strategy Task Analyser** screen, ACI is the Annualised Cost Impact and it quantifies how the strategy task assumptions differ from the actual historical performance. This is a useful number to use to prioritise the review of the strategy task assumptions.

Negative ACI numbers reflect a predicted negative impact on future costs, due to a disconnect between actual performance to date (based on actual work orders) and estimated performance in the future (based on strategy tasks' projection assumptions). In other words, if performance-to-date does not align with the expected performance (based on strategy tasks parameters), there is a disconnection. ACI extrapolates that disconnect in the future and is reflected by negative costs.

For example, a motor life is estimated to be 15,000 hours and costs $100,000 to replace. History shows the motor has an average life of 8,000 hours and costs $150,000 to replace (you're replacing the motor more regularly and at a higher cost than you estimated), then this scenario would be reflected with negative ACI amount.

ACI Field	Description
UOM	Unit of measure of the strategy task.
Actual Occs	Number of actual occurrences of the strategy task based on work order settlement history linked to the strategy task. It does not include occurrences of a parent strategy task if the strategy task is part of a suppression or suppression advanced scheduling group.
SJ	Indicates whether the strategy task is linked to a standard job.
SD	Indicates whether the strategy task has advanced scheduling applied, and if so, is the strategy task a parent or child in the scheduling group. This is an indicator only. Calculations within the grid (such as average actual frequency, average actual cost, and so on) are not affected by this relationship, which is standard for AMT reports. M = Multiple combinations of suppression or dependency in use Sp = Suppression – parent Sc = Suppression – child D = Dependency
ACI Frequency	The additional cost (if a negative number) above the current forecasted cost of the strategy task IF the actual average frequency were to be the basis for strategy task's estimated life in the future. In other words, if you changed the frequency parameter of the strategy task to equal the actual average frequency, the projected future cost of the strategy task would increase by the ACI frequency value.

ACI Field	Description
	Note: this statistic is based on work order settlement history linked to the strategy task.
ACI Cost	The additional cost (if a negative number) above the current forecasted cost of the strategy task IF the actual average cost were to be the basis for strategy task's estimated job cost in the future.
ACI Total	Combined impact of ACI cost and ACI frequency (that is, ACI Cost + ACI Frequency = ACI Total) on future costs.
ALI Labour	The additional labour hours (if a negative number) above the current forecasted labour hours of the strategy task IF the actual average labour hours were to be the basis for strategy task's estimated labour hours in the future.
ALI Total	Combined impact of frequency and labour (that is, the amount of additional labour hours in future based on the actual average frequency and actual average labour hours).
ADI Duration	The additional duration (if a negative number) above the current forecasted duration of the strategy task IF the actual average duration were to be the basis for the strategy task's estimated duration in the future.
ADI Total	Combined impact of frequency and duration (that is, the amount of additional labour hours in future based on actual average frequency and actual average duration).
Proj vs. Actual / UOM Variance	Variance per UOM of the existing projected amount vs. the hypothetical projected amount based on actual averages.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

8. Labour Analysis

You can use the Strategy module to forecast expected labour requirements (based on labour data in the strategy task), and compare estimated to actual, where actual is captured on work orders.

The labour elements stored in strategy tasks are:

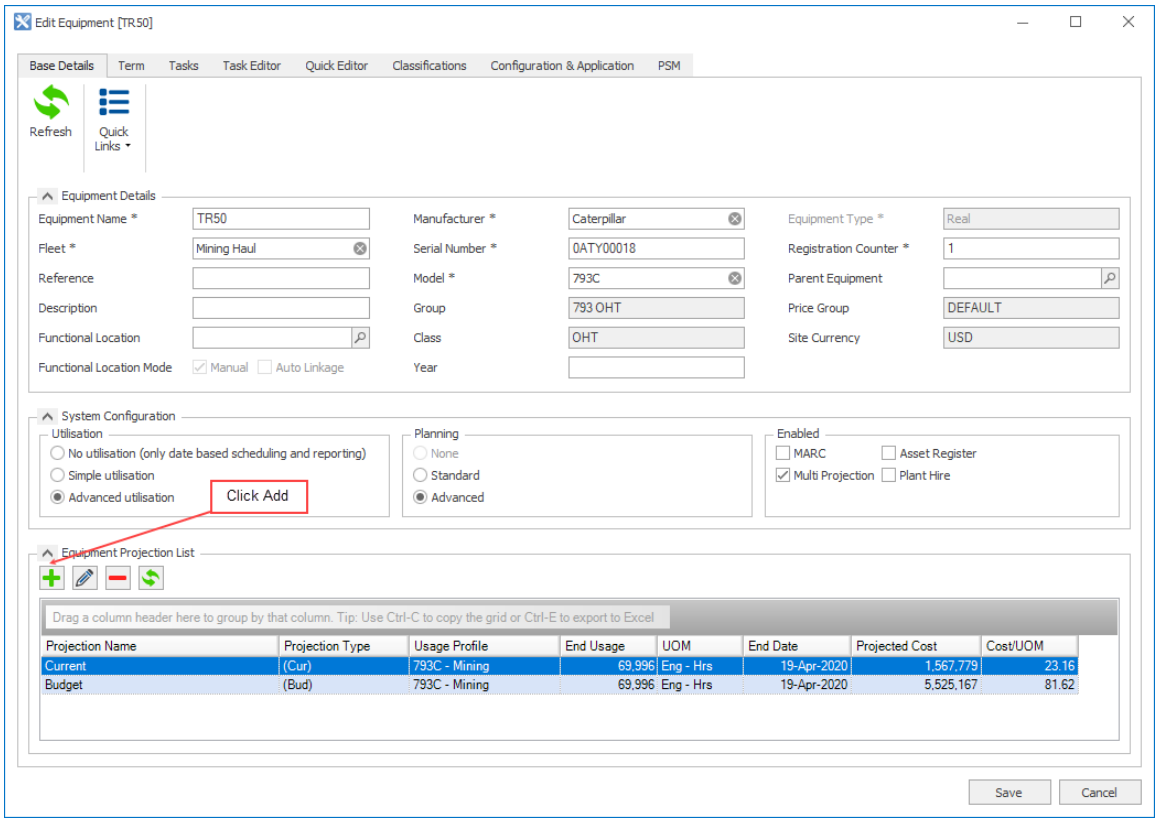
- Labour hours
- Labour activity
- Labour type
- Labour split
- Labour travel cost

9. What-If Analysis

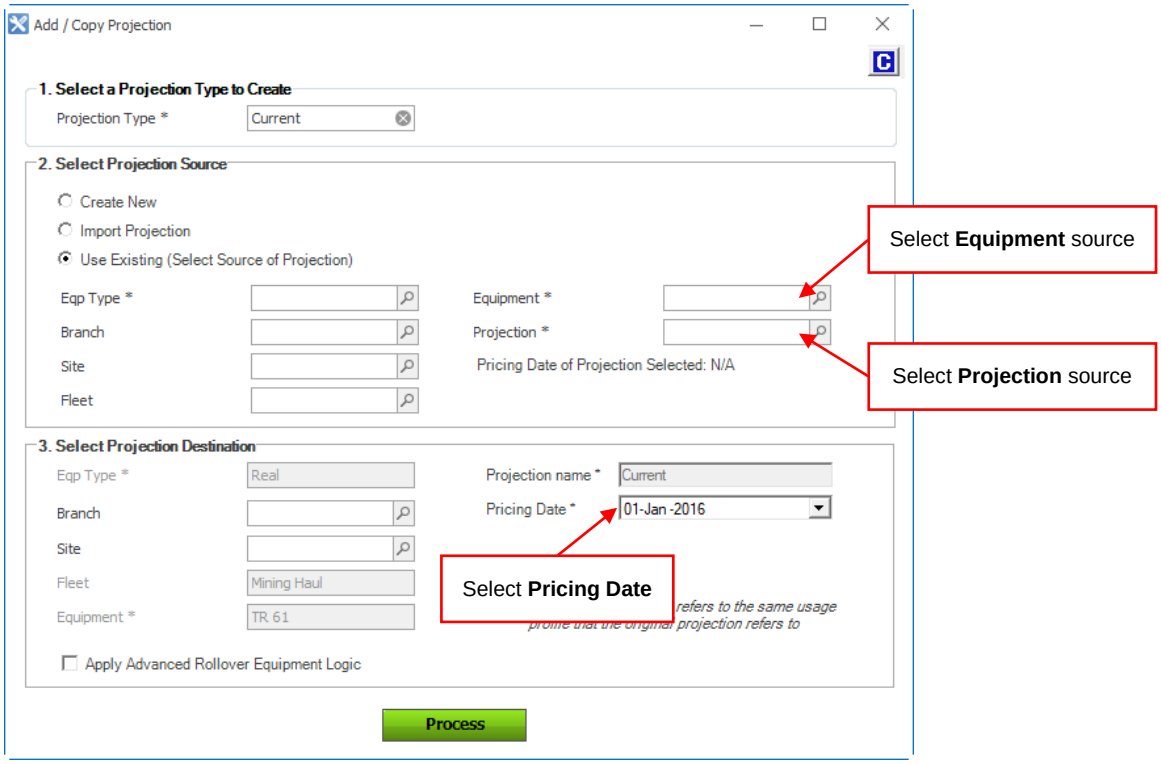
Procedure - Copy What-If to Current Projection

Depending on the amount of changes made to the What If projection, it may be easier to copy the what if projection over the existing current projection of each equipment using the **Add / Copy / Delete Projection** function.

1. Under **Setup & Admin** > **Equipment**, double-click **Equipment Editor**.
2. Apply the filter and click **Add Projection**.



3. Select a **Projection Type** to create.
4. Select a **Projection Source**.
5. If required, change the pricing date.
6. Click **Process**.



You only need to change the pricing date for the new current projection if the projected costs of each strategy in the alternate projection are to be changed to a new date. This action does not adjust the pricing date of strategy tasks linked to standard jobs.

You can escalate or de-escalate the projected costs of each strategy task in the new current projection based on the pricing date entered in the previous step; or leave the pricing date as is.

This action does not adjust the pricing of strategy tasks linked to standard jobs.

You can make the same changes to the current projection as you did to the what if projection (for example, by using the editors). Using this approach, make incremental changes to the current projections.

Unless pricing parameters were changed in the copy process, the current projection's cost/UOM should balance to the what if's cost/UOM.

This process first deletes the current projection of each of the equipment selected and then copies into it the selected projection (in this scenario a what if projection). Then AMT tries to relink work order settlement history to strategy tasks in the new projection. Work order settlements with codes that don't match a strategy task in the new projection are allocated as unassigned costs.

This process cannot be reversed.

It's a good idea to archive the current projection before replacing it, in case a mistake is made in copying the new projection over the existing current projection. After you've successfully copied the new projection, you can delete the archive projection.

Learning Check



Check your understanding of the workshop and ask any related questions.

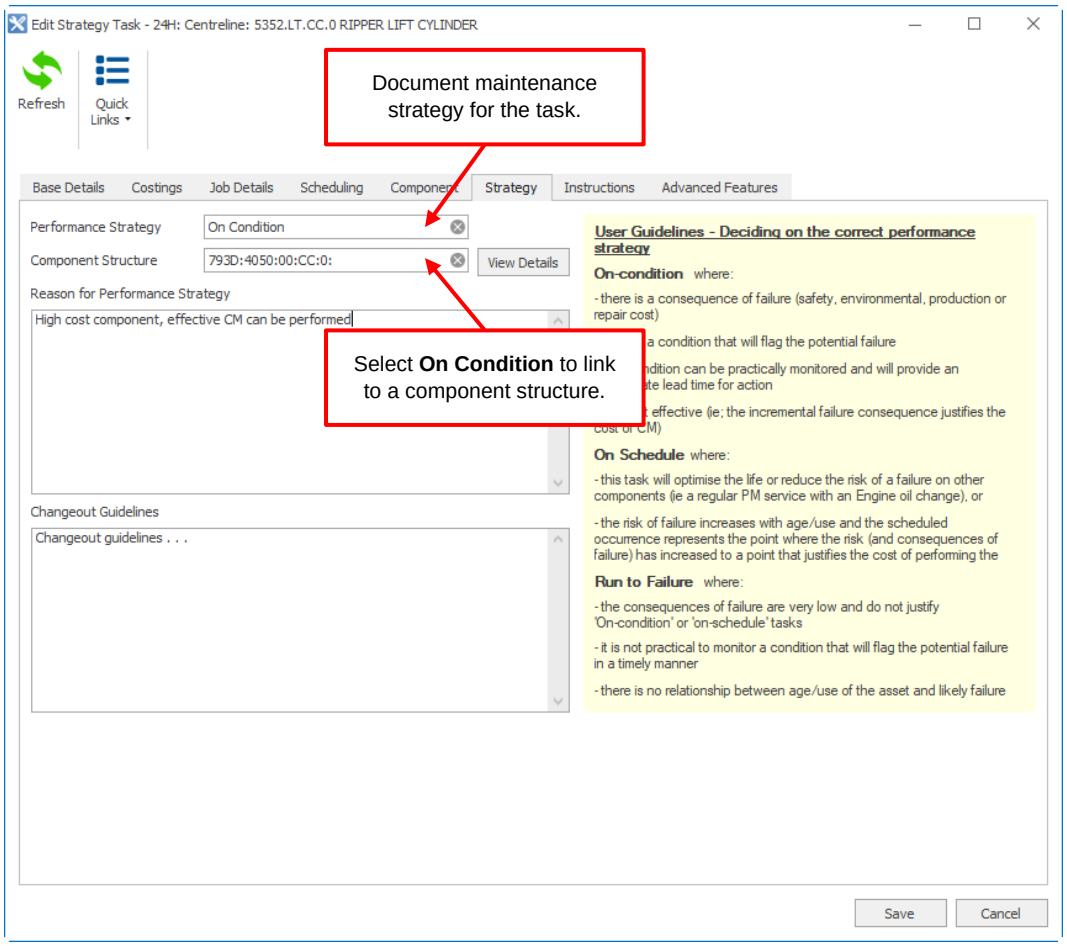
Notes

10. Document Strategy Changes

Following best practices, you should document and store any changes to the strategy – especially long-term adjustments.

Documentation is required for both audits and for sharing information about the strategies with other sites in your organisation. New staff also need to understand the reasons behind the current strategy and why it was modified over time.

Strategy changes for a strategy task can be documented within each strategy task.



We also recommended that your company has a standard format for requesting and documenting strategy changes, particularly long term adjustments, and that these documents are uploaded to the AMT Library.

Create a special document type for the library, for example Strategy Reviews.

For more information on the Library, refer to the **AMT Library** training manual.

11. Monitoring Process

Now that the new strategy is in place, make a copy of it so that future performance can be reviewed against it. To make a copy, create an archive projection.

12. Measuring Performance

On an ongoing basis (such as monthly or quarterly), comparisons can be made between actual performance and the expected position based on the strategy review. You can also compare against the original budget, as it's just another type of projection.

The reports listed below can be used for comparisons:

AMT Report	Description
Budget v Projection Details	Even though this report is called Budget v Current, you can compare any two projections to see differences in strategy task parameters (cost and frequency) and forecasted cost.
Actual v Archive Period Analysis	Even though this report is called Actual v Archive, you can compare any two projections to see the differences in costs per period (month, quarter, year)
What If – Costs Analysis	This text report is similar to the Actual v Archive report. However, instead of breaking down costs by time (such as month, quarter, year) it breaks down costs into cost categories (for example, equipment, component, system) over a specified date range.
Various Analysis Worksheet	Use this text report to analyse the causes of cost variances between the what if projection and a second projection
PCV Chart	Provides a high-level graphical snapshot of actual performance to date versus budget to date and projected.

13. Strategy Reports

On a regular basis (such as monthly or quarterly), the results of the strategy optimisation review and the continuous improvement initiatives (Continuous Improvement Plan) should be reported to management. The reports should detail:

- Key issues identified and the potential impact if they are not addressed.
- Who is responsible for the issue.
- Status of the investigation and corrective action.

For continuous improvement initiatives registered in EM Issues, the **EM Issues List** report provides a summary of all issues.

Below is a summarised list of reports available in **Strategy Optimisation**.

Strategy Definitions
Maintenance Strategy Summary
Failure Mode Details
Life Cycle Analysis
PCV Chart
Life Cycle Cost Summary
PCV Analysis
Variance Analysis Worksheet
Budget vs Current Projection Details
Project Comparison
Cost Analysis
Cost Analysis – “Life-cycle”
Cost Analysis – “To date”
Unassigned Cost Analysis
Analysis of Project Assumptions
Cost Analysis – Life-cycle
Labour
Labour Hours Analysis
Labour Hours Variance Analysis

Component Risk
High Opportunity
High Risk
Component Reliability Analysis
Technical Performance Analysis
Technical Performance Analysis Chart
Technical Performance Details
Component Life Distribution
Component History
Failure Indicators Analysis
Trending
Parts Usage Analysis
Parts Usage Analysis – Qty
Parts Usage Analysis – Value
Parts Usage Details
Parts Usage Analysis – Qty – Global
Parts Usage Analysis – Value – Global
Parts Usage Detail – Global
Part Usage Interrogation – Global
What-if Analysis
What-if – Life Cycle Analysis
What-if – Period Analysis
What-if – Cost Analysis
What-if – Cost per UOM
Variance Analysis Worksheet

Strategy Admin KPIs
Standard Job Coverage
Standard Job Interrogation
WorkScope Coverage
Archiving
Archive Cost Worksheet
Archive Cost Analysis
Actual vs Archive Analysis
Actual vs Archive Period Analysis
Variance Analysis Worksheet
Archive – Costs/UOM

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes